BUILDFEST 2026 / INTERACTIVE WORKSHOP

Making Sense of Data

From Questions to Insights

01
Ask
Define key questions

02
Visualize
Map data patterns

03
Verify
Test hypotheses

04
Decide
Drive action



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TODAY'S CHALLENGE

Your goal: make a decision, not just a dashboard

By the end, you can explain and defend one useful insight.

01 ASK

Start with a useful question.

02 CHECK

Choose a chart and challenge the evidence.

03 ACT

Recommend a cautious next step.

No coding required • Participate through Zoom chat

BUILDFEST 2026 / MEET YOUR FACILITATOR

Hello, I’m Dr. Fardina Alam.

University of Maryland, College Park



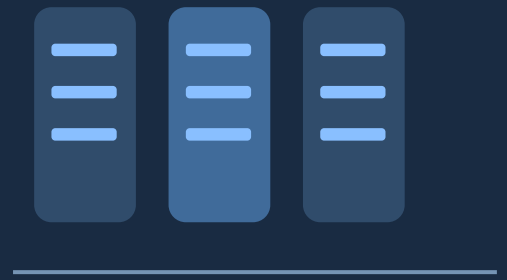
COMPUTER SCIENCE	Faculty member, Department of Computer Science
DATA SCIENCE	Faculty Director, M.S. in Data Science, CMNS
COMMUNITY	Faculty Advisor, Hack4Impact UMD

“No matter your background, you can learn to ask better questions, read data critically, and turn insights into action.”

What is data visualization?

Turning data into charts and graphs that reveal patterns, trends, and comparisons.

DATA SCIENCE



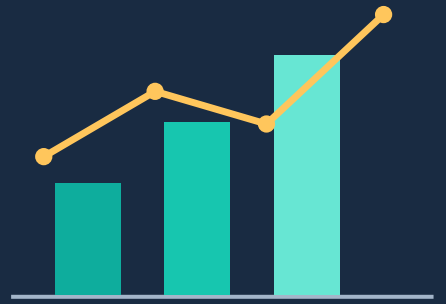
Ask → Collect → Analyze → Act

EXPLORATORY ANALYSIS



Find patterns and outliers

DATA VISUALIZATION



Communicate insights clearly

EXAMPLE 120 survey responses → Compare satisfaction → Identify what needs improvement

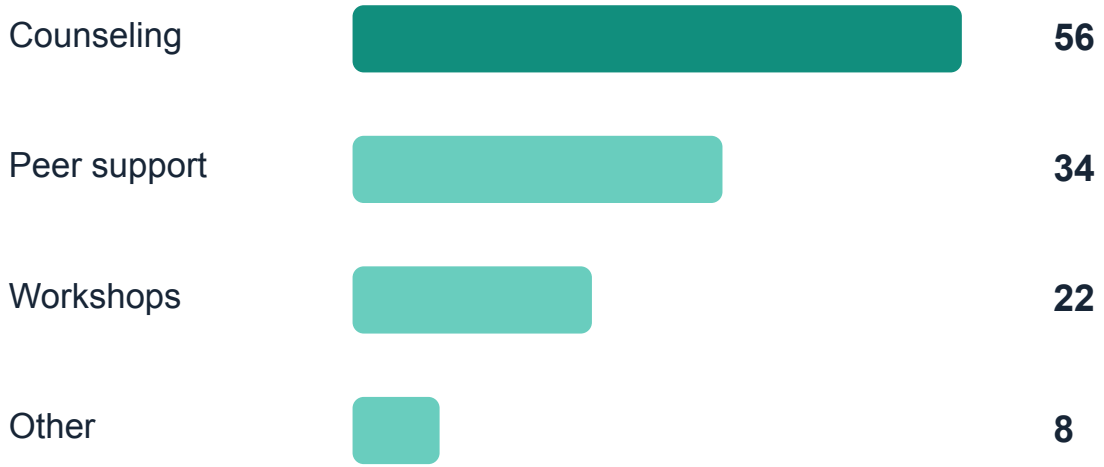
A table makes you read. A chart helps you see.

A fictional non profit survey example:

RAW COUNTS

Counseling	56
Peer support	34
Workshops	22
Other	8

VISUAL COMPARISON



Same fictional survey results. Which format lets you compare categories faster?

Visualization reduces the effort needed to notice patterns—not the need to check them.

THE BIG IDEA

Start with the QUESTION—not the chart.



"A beautiful graph answering the wrong question is still the wrong graph."

COMPARE
Which service is most requested?

DISTRIBUTION
How are ratings spread out?

CHANGE
Are requests increasing monthly?

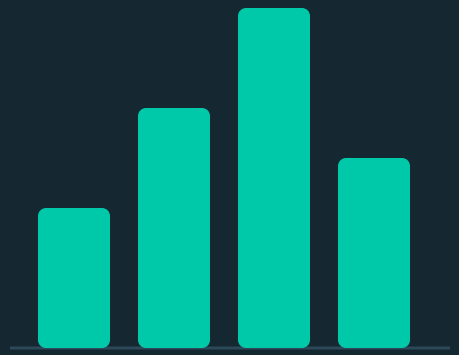
RELATIONSHIP
Are wait times linked to ratings?

Four chart types. Four jobs.

Simple rules that work in Excel, Tableau, Python, or AI-generated charts.

BAR

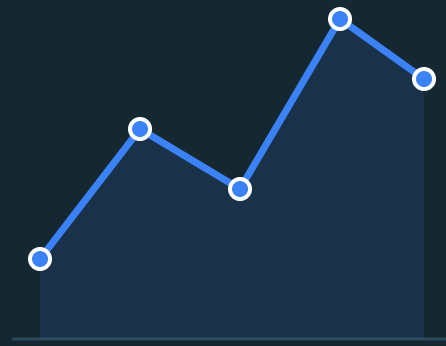
Compare categories



e.g., Service requests

LINE

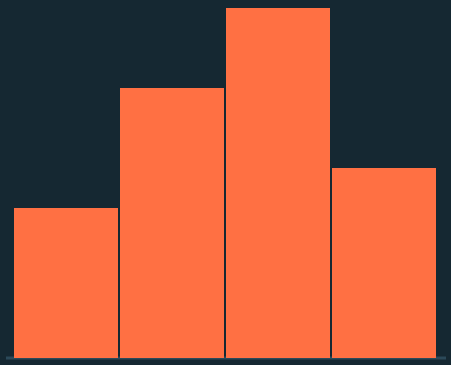
Track change over time



e.g., Monthly attendance

HISTOGRAM

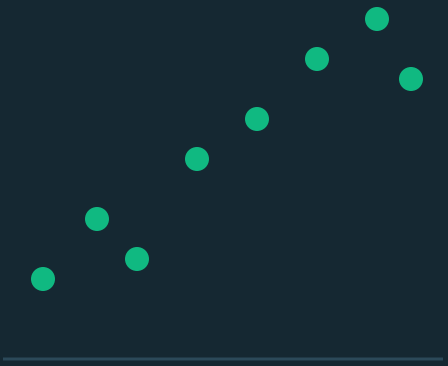
Show numeric distributions



e.g., Wait times

SCATTER

Explore relationships



e.g., Wait time vs rating

AI + Data Visualization Tools: Better Together

AI can support the tools you already use—not replace them.



Excel / Google Sheets

Ask AI to clean data, write formulas, or create charts.



Tableau / Power BI

Use AI to plan dashboard layouts, recommend charts, and explain findings.



Python / R

Ask AI to generate chart code, explain errors, or improve visualizations.



ChatGPT / Claude

Use AI to organize feedback, suggest visualizations, and question conclusions.

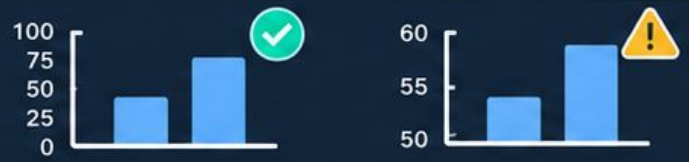


You choose the question and verify the results. AI helps with the work.

ESSENTIAL RULES

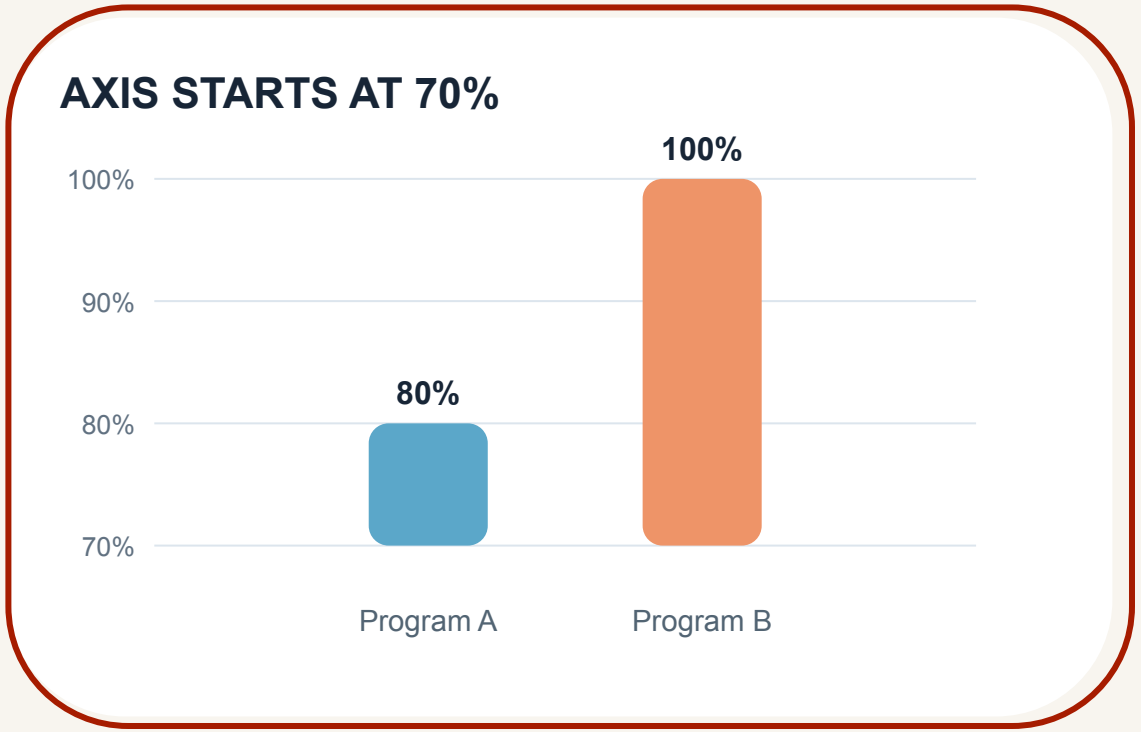
The five checks before sharing any chart

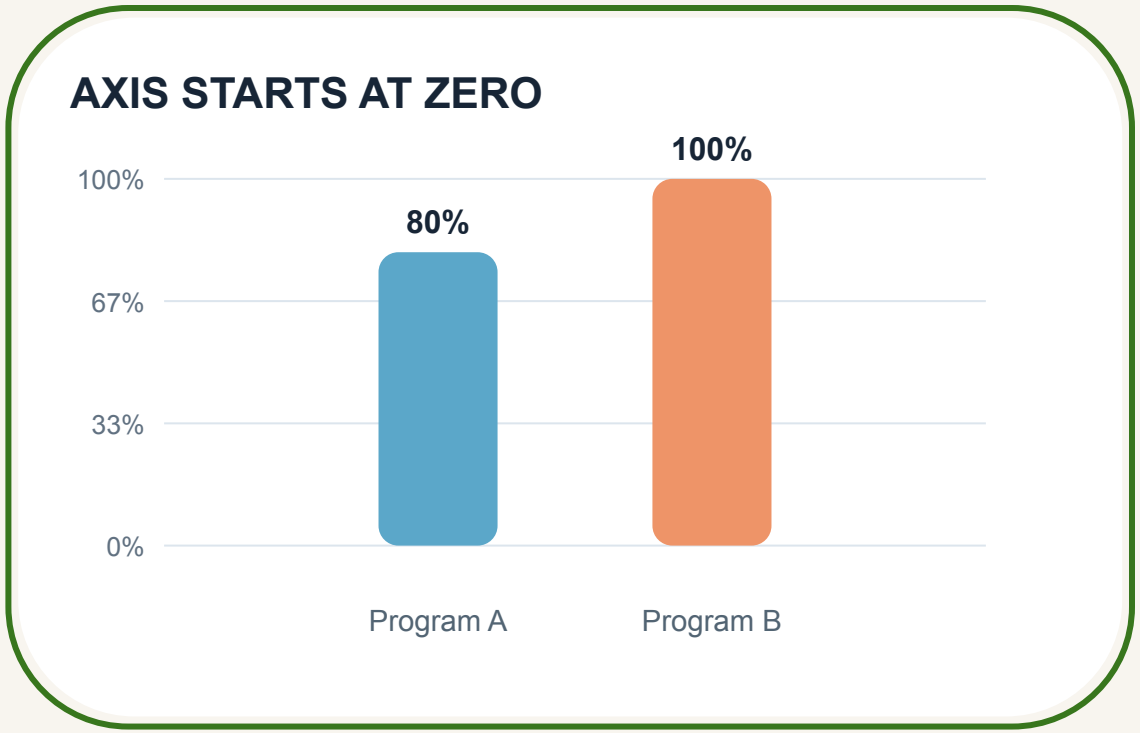
Use these to critique human-made AND AI-generated charts.

01	MATCH		Does the chart answer the question?
02	LABEL		Are labels, units, and sources clear?
03	SCALE		Is the scale misleading?
04	DENOMINATOR		Out of how many? Who is missing?
05	HONESTY		Do the conclusions match the data?

Same data. Very different impression.

Both charts show Program A = 80% and Program B = 100%.





A truncated bar axis can exaggerate a difference. Always read the numbers AND the scale.

INTERPRETATION FRAMEWORK

A finding is not yet a recommendation.

Move carefully from observation to meaning to action.

OBSERVATION

"60 out of 100 people reported satisfaction."



INTERPRETATION

"Many were satisfied, but we need more information."



ACTION

"Review concerns before expanding."



WHAT WE SEE → WHAT IT MAY MEAN → WHAT TO DO NEXT

NEXT: INTERACTIVE CASE

INTERACTIVE CASE

Welcome to the Insight Rescue Room

You are advising a fictional nonprofit that surveyed 100 people.

DASHBOARD UNDER REVIEW

REPORTED SATISFACTION

67%

of ALL participants

PROGRAM CONCLUSION

Proven successful!

Expand immediately

Does this story match the evidence?

HOW TO PARTICIPATE

Type A/B/C or a short answer in Zoom chat. No breakout rooms or coding.

SPOT THE PROBLEM → CHAT → DISCUSS

What does our survey look like?

Fictional nonprofit • **100** respondents • **Preview of 5 responses**

Respondent	Service used	Wait (days)	Satisfaction
001	Counseling	12	Dissatisfied
002	Peer support	3	Satisfied
003	Workshops	1	Satisfied
004	Counseling	15	Dissatisfied
005	Peer support	7	No response

Each row is one person. What useful question could we ask from these columns?

Illustrative sample only—not the full dataset. Answers to one question may be missing.

NEXT


MISSION 01
Ask the Right Question
Your first step to finding meaningful insights

MISSION 1 • 3 MINUTES

Mission 1: Which question should we investigate first?

Goal: understand what might be linked te dissatisfaction.

A Which service was selected most often?

B Are longer waits linked to lower satisfaction?

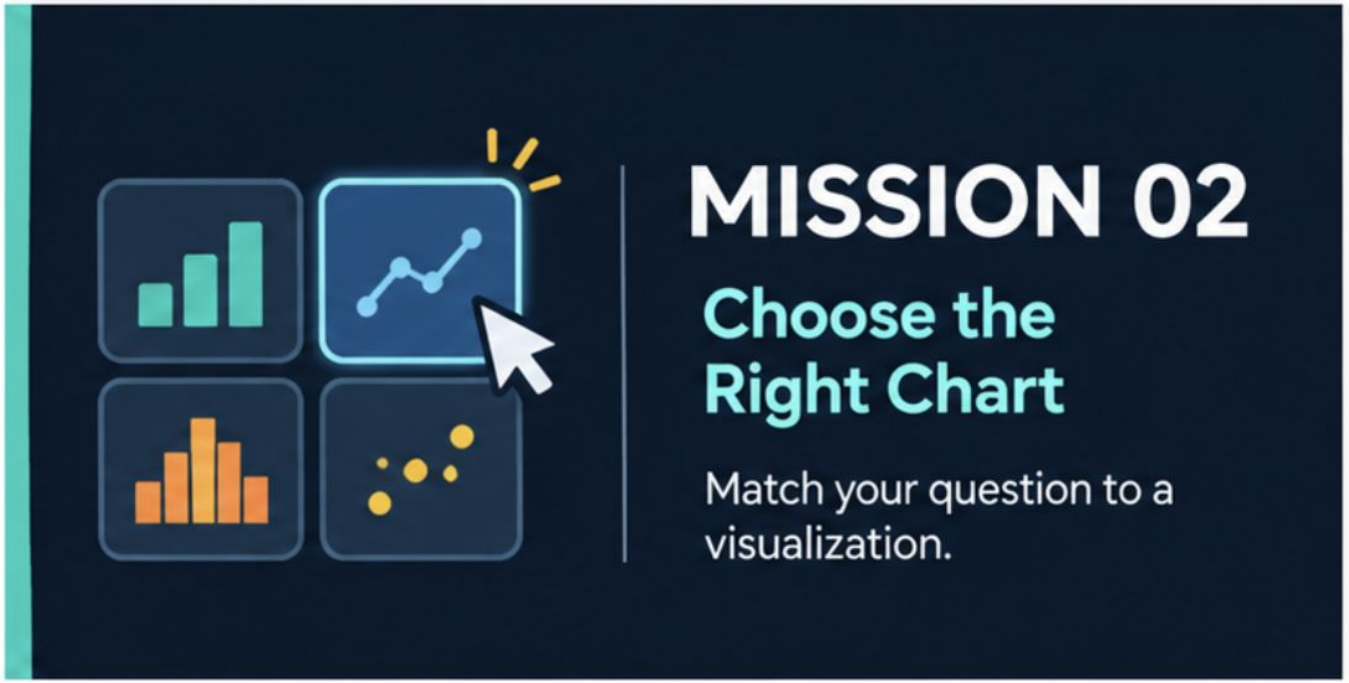
C What percentage of respondents were satisfied?

Sample: 5 of 100 fictional survey responses

ID	Service used	Wait (days)	Satisfaction
001	Counseling	12	Dissatisfied
002	Peer support	3	Satisfied
003	Workshops	1	Satisfied
004	Counseling	15	Dissatisfied
005	Peer support	7	No response

CHAT: A, B, or C? Which helps us investigate dissatisfaction?

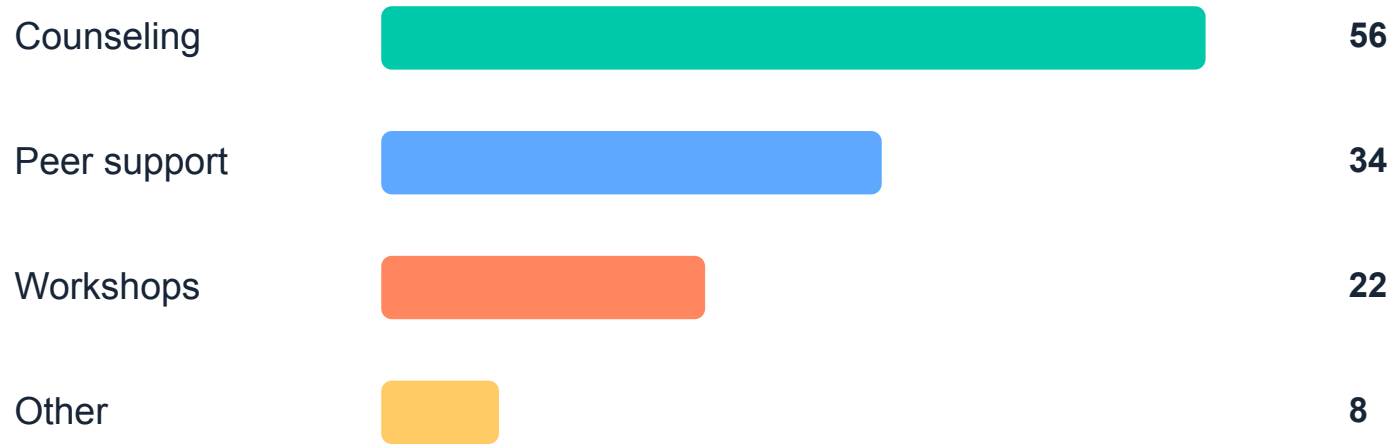
NEXT


MISSION 02
Choose the Right Chart
Match your question to a visualization.

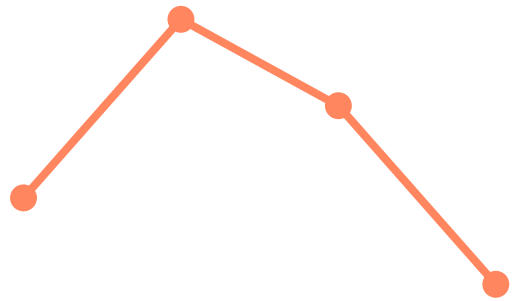
Mission 2: Chart battle

Question: which service was selected most often?

OPTION A • BAR CHART



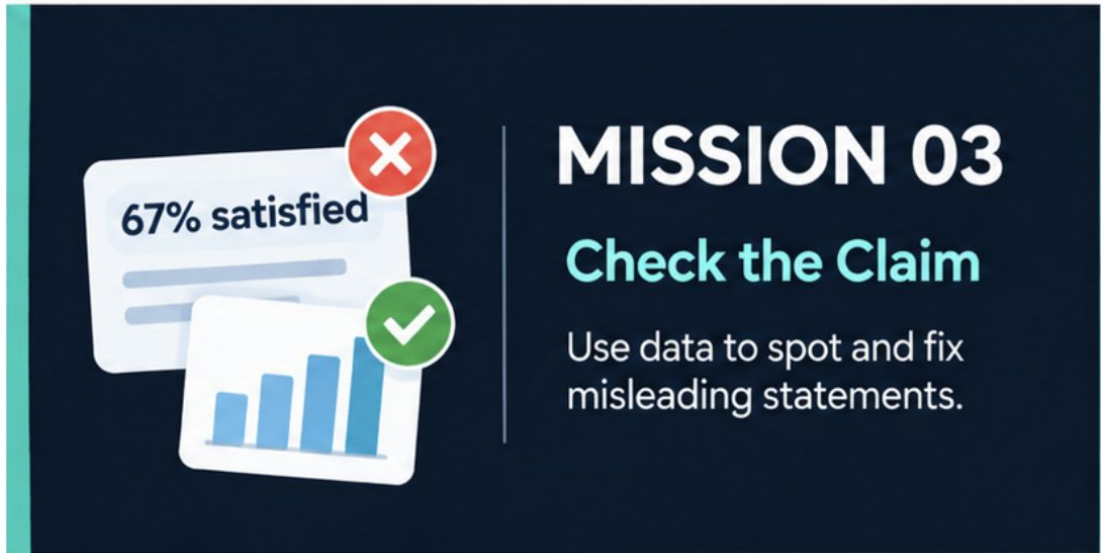
OPTION B • LINE



A line implies an order that categories do not have.

CHAT: A or B—which is more appropriate, and why?

NEXT


67% satisfied
MISSION 03
Check the Claim
Use data to spot and fix misleading statements.

Mission 3: AI on trial

First, let humans critique a confident-sounding AI summary.

“AI-GENERATED” DRAFT

“67% of ALL participants were satisfied. This proves the nonprofit’s services work. Expand the program immediately.”

SOURCE DATA

Satisfied	60
Dissatisfied	30
No answer	10
Total surveyed	100

CHAT: What is wrong? Try to identify ONE problem.

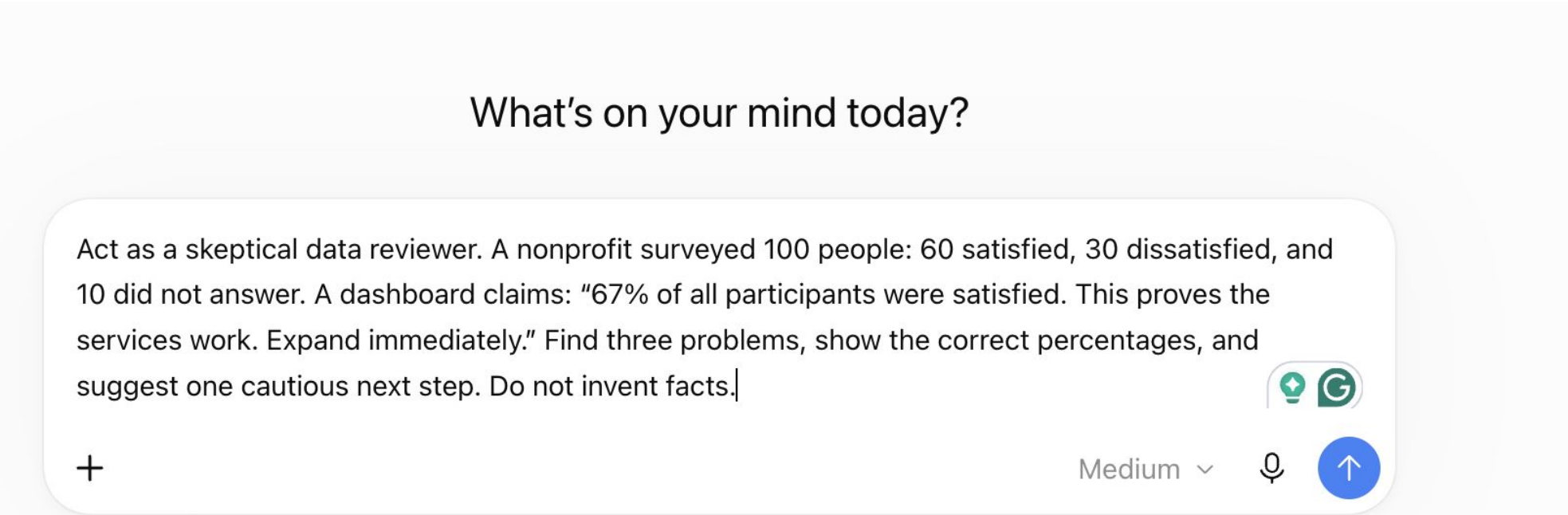
Prompt Duel: make AI your skeptical reviewer

COPY INTO CHATGPT OR CLAUDE

Act as a skeptical data reviewer. A nonprofit surveyed 120 people: 72 satisfied, 30 dissatisfied, and 18 skipped the question. A dashboard claims: “70.6% of all respondents were satisfied, proving our services work; expand immediately.” Identify every unsupported or misleading part. Show both correct denominators. Rewrite the insight and propose ONE cautious next step. Do not invent facts.

HUMAN FIRST → AI SECOND → VERIFY THIRD

Prompt in ChatGPT



ChatGPT Responses

Act as a skeptical data reviewer. A nonprofit surveyed 100 people: 60 satisfied, 30 dissatisfied, and 10 did not answer. A dashboard claims: “67% of all participants were satisfied. This proves the services work. Expand immediately.” Find three problems, show the correct percentages, and suggest one cautious next step. Do not invent facts.

Three problems with the dashboard claim

1. **Wrong percentage:** 67% is based only on the 90 people who answered, not all 100 participants.
2. **Unproven conclusion:** Satisfaction does not prove the services are effective.
3. **Rushed recommendation:** The data alone does not justify expanding services immediately.

Correct percentages

Response	Out of all 100	Out of 90 who answered
Satisfied	60%	66.7%
Dissatisfied	30%	33.3%
Did not answer	10%	Excluded

One cautious next step

Investigate why 30 people were dissatisfied and why 10 did not answer before deciding whether to expand.

Better dashboard statement: "60% of all surveyed participants reported satisfaction. More information is needed before recommending expansion."

Reveal: check the denominator

Two different percentages answer two different questions.

SOURCE DATA	
Satisfied	60
Dissatisfied	30
No answer	10
Total surveyed	100

60 ÷ 100 = 60%

Of everyone surveyed

60 ÷ 90 = 66.7%

Of people who answered

Better statement: ***"Of the 90 people who answered, 60 (66.7%) were satisfied. The other 10 people did not answer the question."***

KEY LESSON

AI can help—but you still make the judgment

Use AI to find weaknesses in your interpretation, not to manufacture confidence.

ASK

What could be wrong with this claim?

VERIFY

Recalculate numbers from the source.

REVISE

State uncertainty and avoid overclaiming.

Useful prompt habit: “Challenge this interpretation using only the data provided.”

NEXT


FINAL CHALLENGE
Make Your
Recommendation
Turn what you found in the data into
a clear, cautious next step.

FINAL CHALLENGE • 4 MINUTES

Your turn: make a safer recommendation

One sentence about what the data shows + one cautious next step.

60 satisfied • 30 dissatisfied • 10 no answer

Complete this in Zoom chat:

“We found _____. Before expanding, we should _____.”

THE DATA-TO-DECISION PLAYBOOK

From Data to Action: Five Simple Steps

Use them with spreadsheets, coding, dashboards, and AI.

RECAP

01 ASK



Start with a decision-relevant question.

02 SHOW



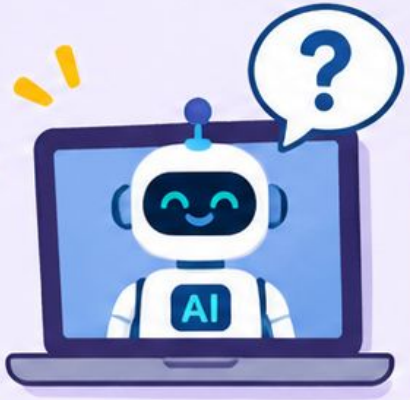
Choose a visualization that fits.

03 CHECK



Read the scale, labels, and denominator.

04 QUESTION AI



Ask for critique; verify every claim.

05 ACT



Recommend only what the evidence supports.

QUESTIONS & DISCUSSION

What makes an insight trustworthy?

Thank you • BuildFest 2026

A clear question. An honest chart. A defensible conclusion.

Questions? Or share one thing you will check before trusting a dashboard.

Thank you for participating!